ZA-GAS Daily Lag-Robustness Test — Belo Horizonte

ZA-GAS Model Comparison — Thesis Working Report

Abstract

This report compares two specifications of the Zero-Augmented Generalised Autoregressive Score (ZA-GAS) model for precipitation time series. In the phi-only specification only the GB2 location parameter φ is time-varying; in the phi-xi specification both φ and the shape parameter ξ follow the GAS score-driven recursion. Models are evaluated at daily and monthly cadences on in-sample fit (log-likelihood, AIC, BIC) and out-of-sample forecast accuracy (CRPS, RMSE, MAD). Probabilistic calibration is assessed via PIT uniformity tests (Kolmogorov-Smirnov) and 95 % coverage tests (Kupiec, Christoffersen).

⚠ **Scope Note**

This is the first experiment in the thesis research. Models were fitted and evaluated only for the Belo Horizonte (BH) time series. Results for other Brazilian weather stations will be presented in subsequent experiments once the full spatial sweep is complete.

Model Descriptions

Framework: Zero-Augmented GAS

The observation equation is zero-augmented. Conditional on information F_{t-1},

$$p(y_t \mid \mathcal{F}_{t-1}) = (1-\pi_t)\mathbf{1}\{y_t = 0\} + \pi_t g(y_t; f_{t|t-1}, \theta)\mathbf{1}\{y_t > 0\}.$$

The positive component $g(\cdot)$ is GB2 with log-link parameters. The state vector $f_{t|t-1}$ contains the time-varying positive-part parameters: either $f_{t|t-1} = (\phi_{t|t-1})$ or $f_{t|t-1} = (\phi_{t|t-1}, \xi_{t|t-1})'$.

The GAS recursion follows the one-step-ahead prediction notation used in score-driven models:

$$f_{t+1|t} = \omega + \sum_{\ell \in L} A_\ell s_{t-\ell+1} + \sum_{\ell \in L} B_\ell f_{t-\ell+1|t-\ell}.$$

Here $s_t = \mathcal{I}_{t|t-1}^{-1}\nabla_t$ is the Fisher-information-scaled score, $\nabla_t = \partial \log p(y_t|\mathcal{F}_{t-1})/\partial f_{t|t-1}$. The score is set to zero for exact-zero observations because those observations identify the zero probability rather than the positive GB2 density.

For the non-seasonal GAS(1,1) special case this reduces to $f_{t+1|t} = \omega + As_t + Bf_{t|t-1}$.

Pi Dynamics

The non-zero probability is $\pi_t = \Lambda(\eta_t)$, where $\Lambda(x) = 1/(1+\exp(-x))$ is the logistic link.

The implemented AR-logistic recursion is $\eta_t = \omega_0 + \rho\eta_{t-1} + \sum_{\ell \in L} \omega_{y,\ell} y_{t-\ell}$.

The same seasonal lag set L is used for the GAS recursion and the π_t dynamics, so the zero/non-zero occurrence process and the positive-part intensity process can both react to short-memory and calendar-seasonal precipitation information.

Seasonal Lag Sets

Daily cadence: L = {1, 2, 3, 364, 365, 366, 367} - seven lags covering +/-1 day around the annual cycle

Monthly cadence: L = {1, 2, 3, 11, 12, 13} - six lags covering +/-1 month around the annual cycle

Model 1 - ϕ-only

Only ϕ is time-varying; ξ is kept at its estimated constant value throughout the sample. This is the parsimonious baseline:

it captures location shifts in positive-part precipitation while holding the spread/shape channel fixed.

Model 2 - ϕ, ξ

Both ϕ and ξ follow the GAS update. This allows both the positive-part scale/location and the GB2 shape channel to react dynamically to new observations, potentially improving calibration in regimes where rainfall intensity and dispersion change together.

Parameter Counts

Daily ['phi']: 10 estimated parameters (4 GAS dynamics, 3 static positive-part, 3 pi-dynamics)

Daily ['phi', 'xi']: 13 estimated parameters (8 GAS dynamics, 2 static positive-part, 3 pi-dynamics)

Estimation

Parameters are estimated by maximising the full log-likelihood using L-BFGS-B with box constraints. Standard errors are Wald-type from the optimizer's inverse Hessian approximation. Confidence intervals are at the 95% level.

Series Diagnostics

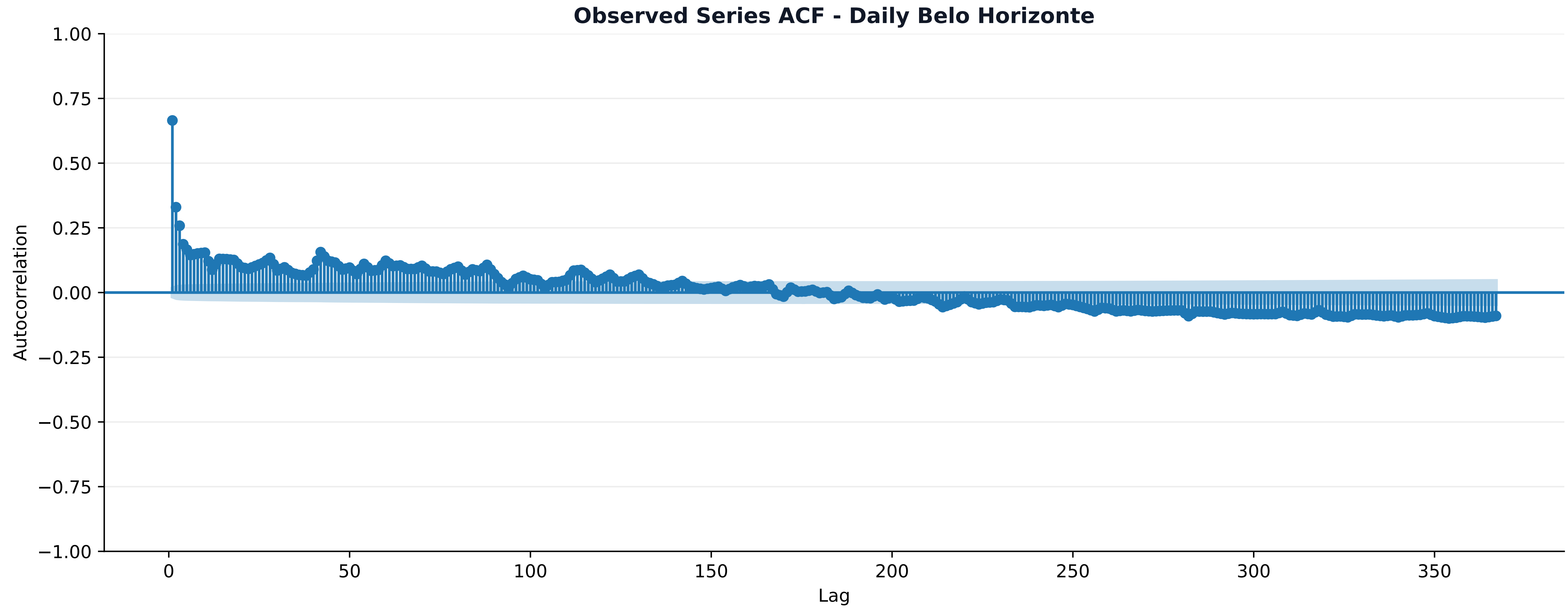
Section Overview

This section describes the observed precipitation series before any model-specific diagnostics are shown. The statistics and ACF charts are about the data series itself, not about a fitted model.

Observed Series Descriptive Statistics

Series	sample	n	min	max	mean	std	prop_zero	q01	q05	q25	q50	q75	q90	q95	q99
Daily Belo Horizo	IS	3652.000	0.000	171.800	4.008	11.630	0.723	0.000	0.000	0.000	0.000	0.500	13.900	25.545	53.349
Daily Belo Horizo	OOS	731.000	0.000	92.800	4.153	10.837	0.706	0.000	0.000	0.000	0.000	0.600	15.000	29.050	50.280
Daily Belo Horizo	Full	4383.000	0.000	171.800	4.032	11.500	0.720	0.000	0.000	0.000	0.000	0.500	14.000	26.470	52.326

Note: Statistics are computed for the raw precipitation series. Full = IS and OOS concatenated.



ACF of observed precipitation, lags 1-367; shaded bands are approximate 95% confidence intervals.

Per-Model Diagnostics

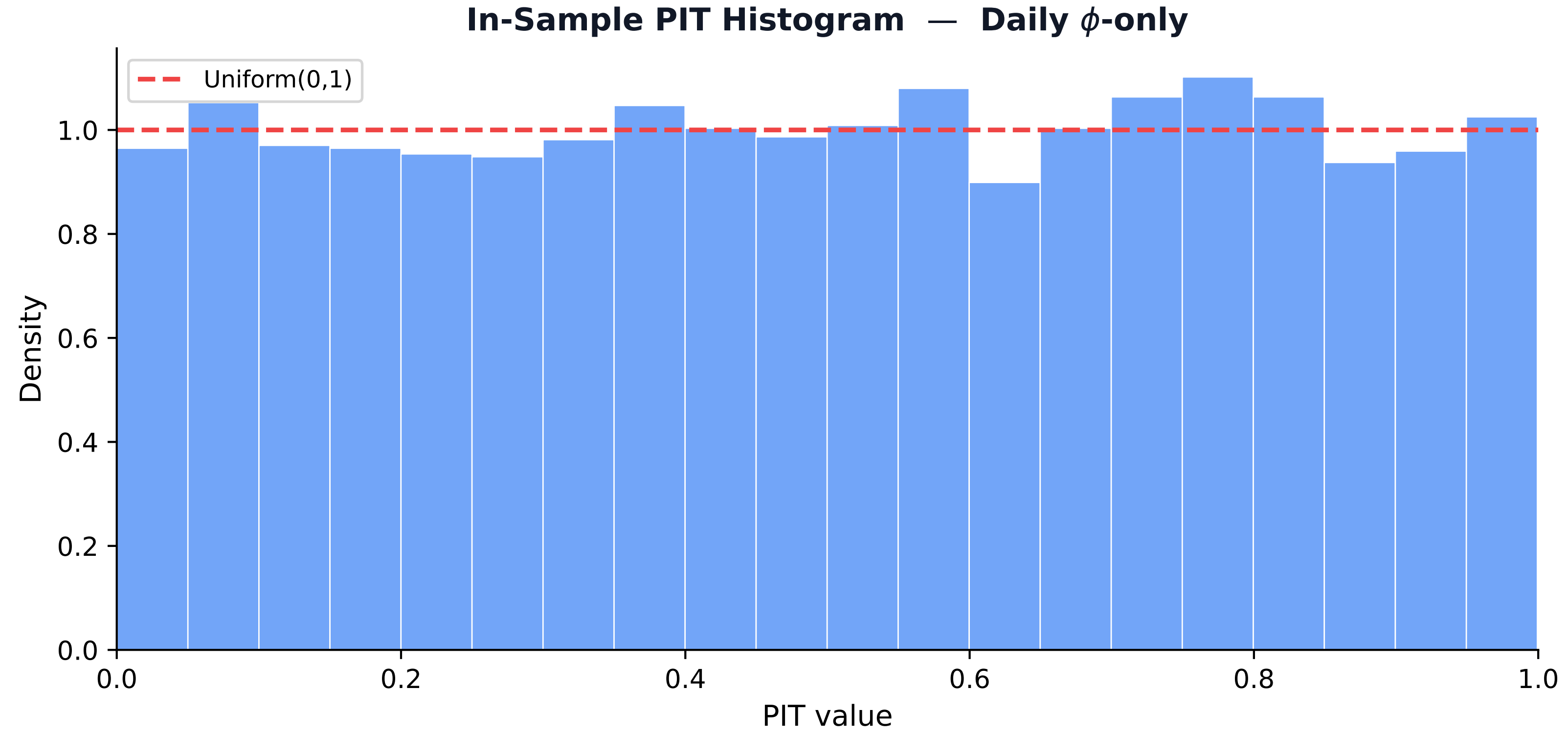
Section Overview

The following pages present individual diagnostic charts and tables for each fitted model: PIT histogram (calibration), quantile residual ACF (serial independence), KS test statistics, and estimated parameters. All diagnostic charts use only in-sample observations. Cross-model comparisons are deferred to the Comparison Tables section.

Diagnostics: Daily ϕ-only

Scope

This section presents in-sample diagnostic charts and tables for the Daily ϕ-only model. These are model-specific checks of probabilistic calibration (PIT) and residual autocorrelation (ACF). Cross-model comparisons are in the Comparison Tables section.

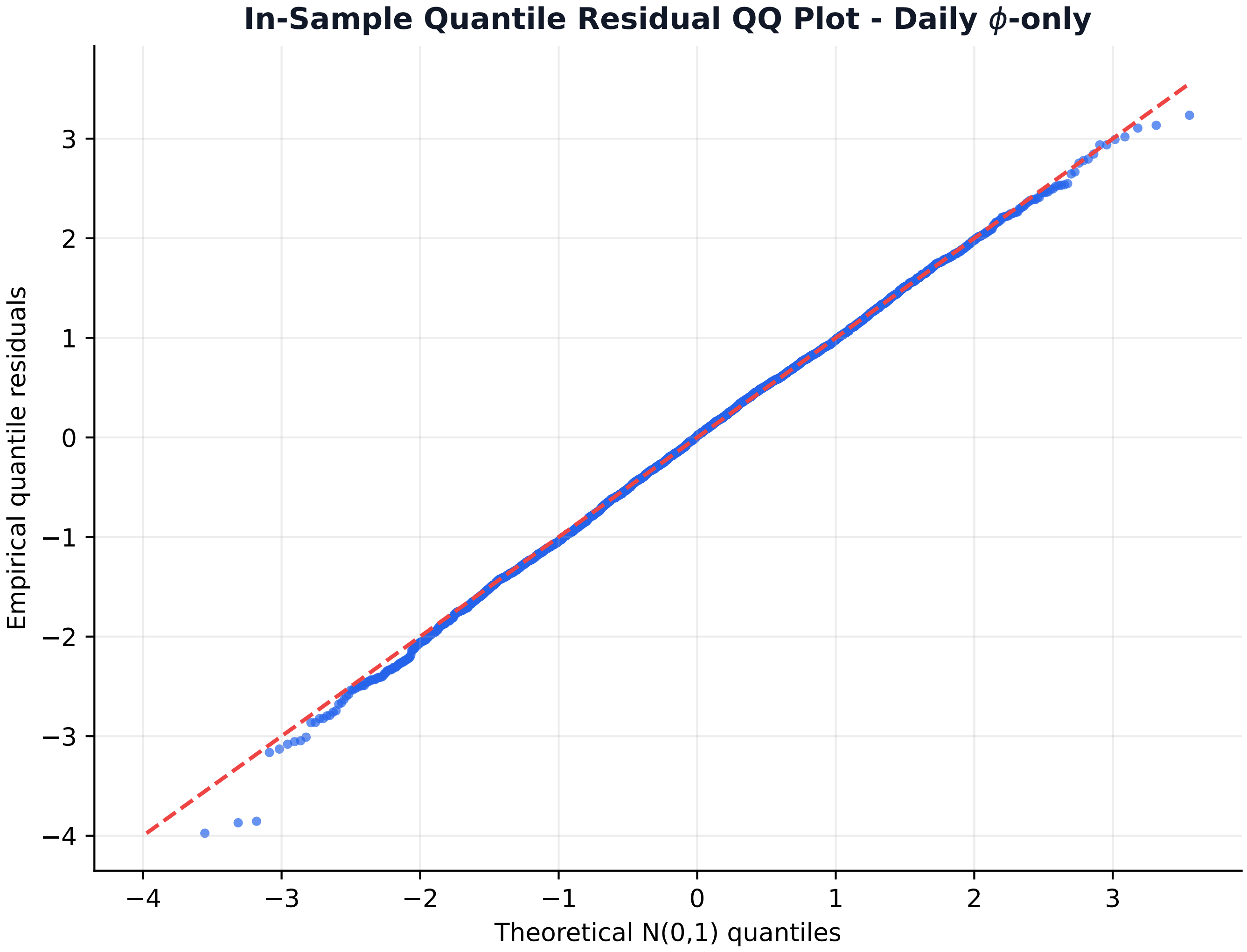


Bars close to height 1 (the red dashed line) indicate good probabilistic calibration.

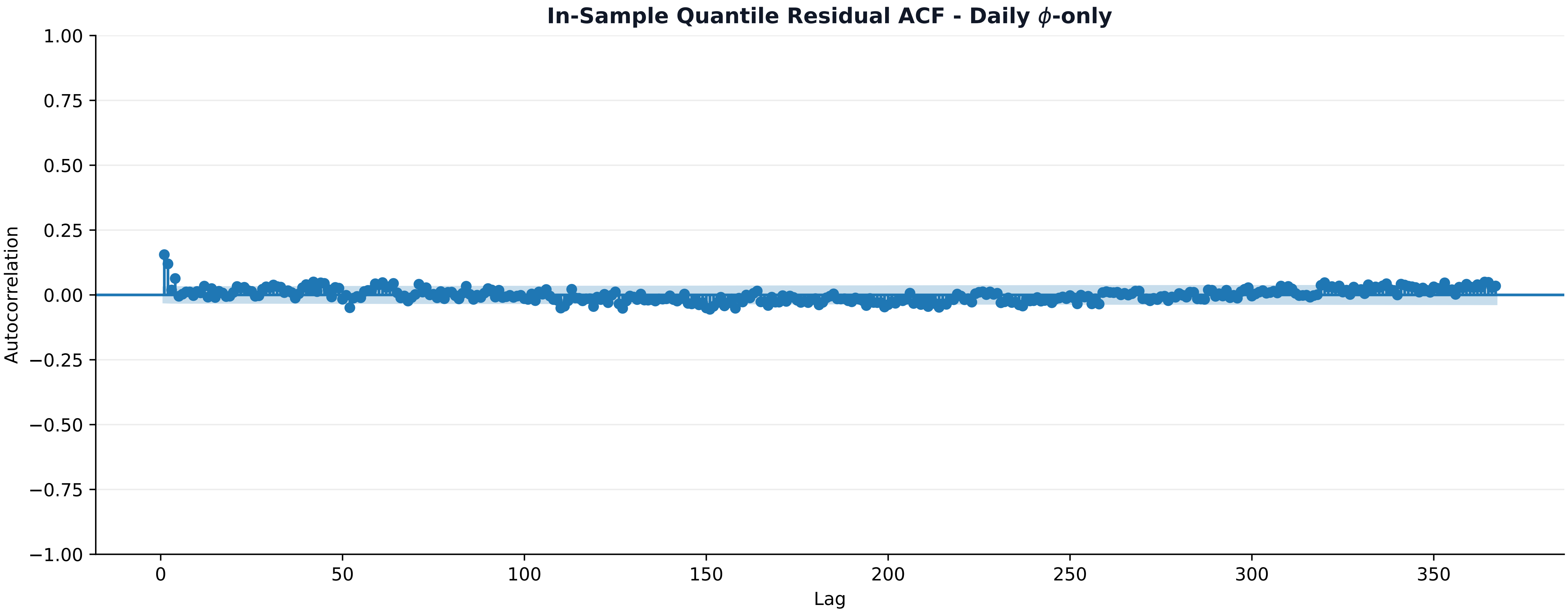
PIT Kolmogorov-Smirnov Test — Daily ϕ-only

sample	n	ks_stat	pvalue	reject_5pct
IS	3651.000	0.012	0.617	0.000
OOS	731.000	0.030	0.518	0.000
Full	4382.000	0.015	0.297	0.000

Note: KS test of PIT values against Uniform(0, 1). A well-calibrated model has p-value > 0.05 and ks_stat close to zero. IS = in-sample; OOS = out-of-sample; Full = concatenated sample.



Points close to the dashed 45-degree line indicate approximately standard-normal quantile residuals.



One model only. Lags 1-367; shaded bands are approximate 95% confidence intervals under white-noise residuals.

Jarque-Bera Test for Quantile Residuals - Daily ϕ-only

sample	n	skewness	kurtosis	JB_stat	pvalue	reject_5pct
IS	3651.000	-0.082	3.017	4.156	0.125	0.000

Note: Jarque-Bera tests whether in-sample quantile residuals are compatible with a Gaussian distribution. Red cells indicate rejection at 5%.

Estimated Parameters — Daily ϕ-only

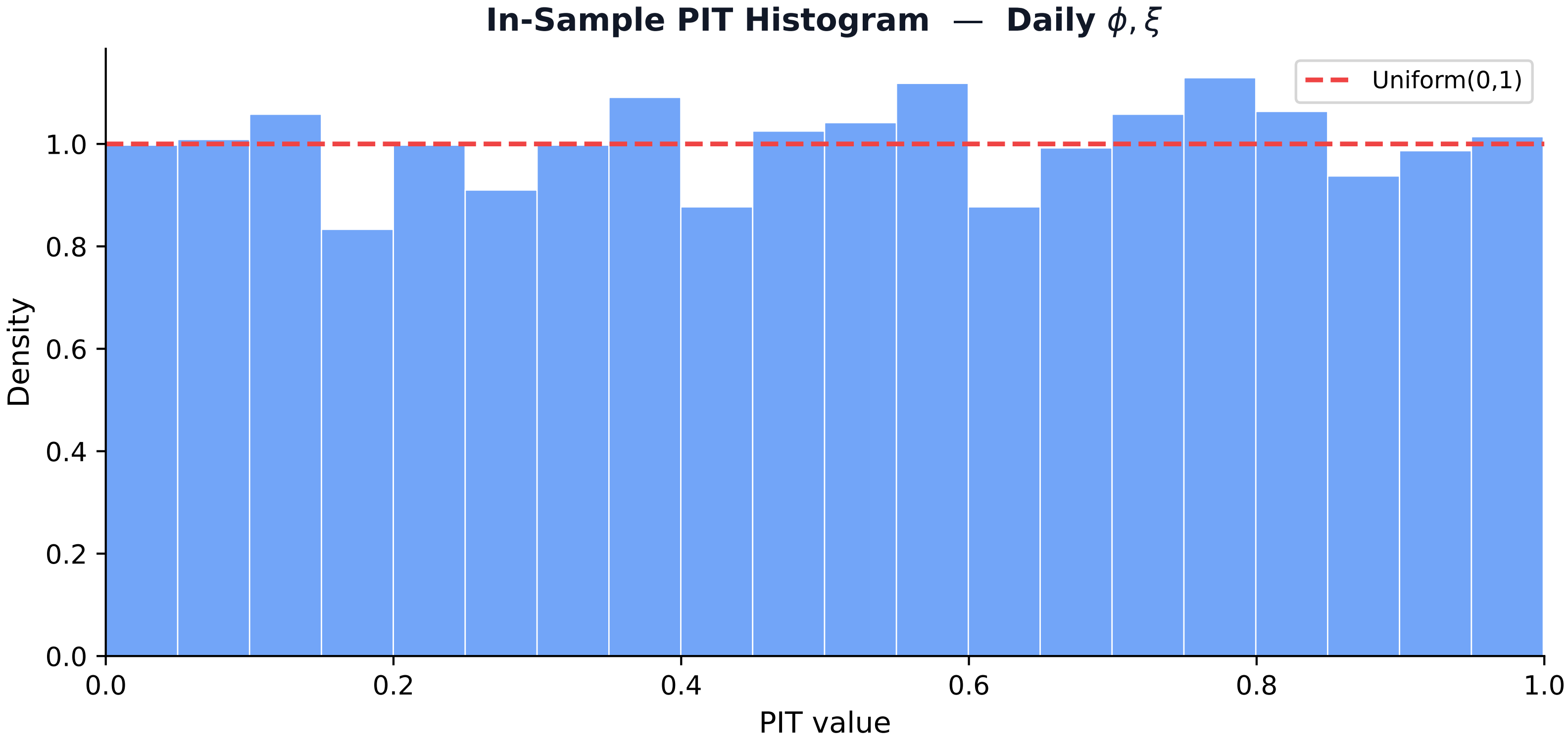
parameter	block	estimate	std_error	ci_lower_95	ci_upper_95
omega_phi	gas	0.3316	0.0520	0.2297	0.4334
f0_phi	gas	11.2383	0.9555	9.3656	13.1109
A_phi_1	gas	0.0713	0.0203	0.0315	0.1111
B_phi_1	gas	0.9686	0.0049	0.9591	0.9782
xi	positive_static	-0.1082	0.0007	-0.1096	-0.1067
gamma	positive_static	0.1745	0.0267	0.1222	0.2268
zeta	positive_static	6.7566	0.0003	6.7560	6.7571
omega0	pi	-0.3912	0.0132	-0.4171	-0.3653
rho	pi	0.7675	0.0107	0.7466	0.7884
omega_y_1	pi	0.0376	0.0018	0.0341	0.0411

Note: Estimates from the optimizer used for the cached model. Standard errors and 95 % Wald CIs from the inverse Hessian; interpret with caution near optimizer bounds.

Diagnostics: Daily ϕ, ξ

Scope

This section presents in-sample diagnostic charts and tables for the Daily ϕ, ξ model. These are model-specific checks of probabilistic calibration (PIT) and residual autocorrelation (ACF). Cross-model comparisons are in the Comparison Tables section.

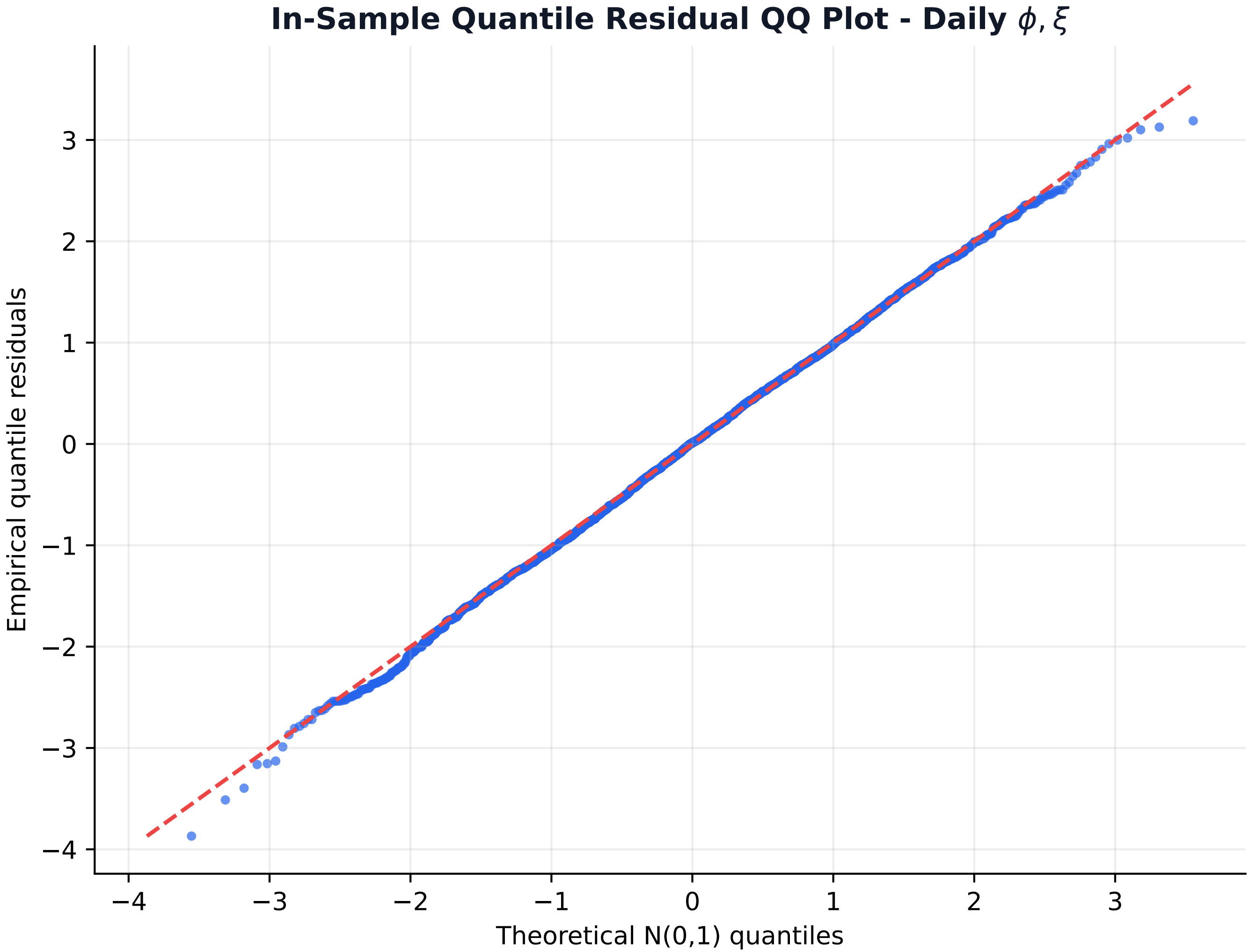


Bars close to height 1 (the red dashed line) indicate good probabilistic calibration.

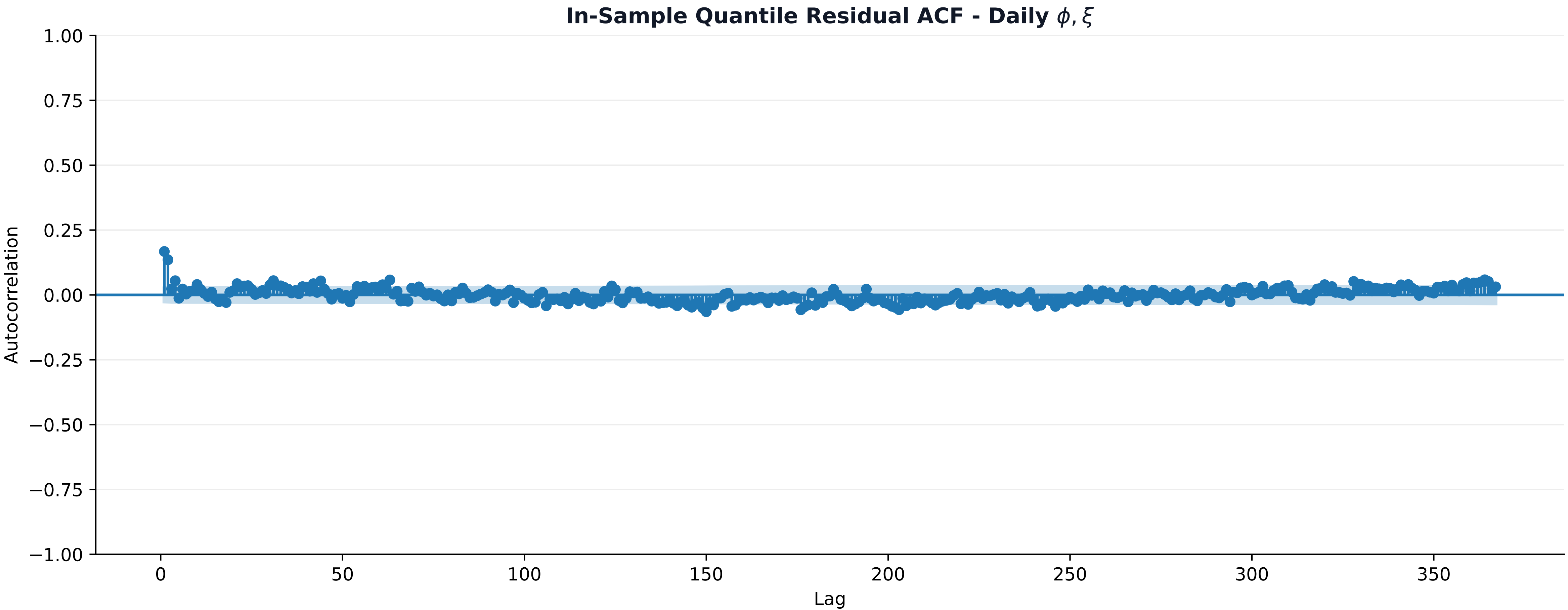
PIT Kolmogorov-Smirnov Test — Daily ϕ, ξ

sample	n	ks_stat	pvalue	reject_5pct
IS	3651.000	0.013	0.528	0.000
OOS	731.000	0.030	0.518	0.000
Full	4382.000	0.015	0.243	0.000

Note: KS test of PIT values against Uniform(0, 1). A well-calibrated model has p-value > 0.05 and ks_stat close to zero. IS = in-sample; OOS = out-of-sample; Full = concatenated sample.



Points close to the dashed 45-degree line indicate approximately standard-normal quantile residuals.



One model only. Lags 1-367; shaded bands are approximate 95% confidence intervals under white-noise residuals.

Jarque-Bera Test for Quantile Residuals - Daily ϕ, ξ

sample	n	skewness	kurtosis	JB_stat	pvalue	reject_5pct
IS	3651.000	-0.053	2.949	2.072	0.355	0.000

Note: Jarque-Bera tests whether in-sample quantile residuals are compatible with a Gaussian distribution. Red cells indicate rejection at 5%.

Estimated Parameters — Daily ϕ, ξ

parameter	block	estimate	std_error	ci_lower_95	ci_upper_95
omega_phi	gas	0.4161	0.0153	0.3862	0.4460
f0_phi	gas	13.6353	0.0319	13.5728	13.6979
A_phi_1	gas	0.0619	0.0080	0.0463	0.0775
B_phi_1	gas	0.9669	0.0012	0.9645	0.9693
omega_xi	gas	-0.0028	0.0010	-0.0048	-0.0008
f0_xi	gas	-0.3702	0.0424	-0.4533	-0.2871
A_xi_1	gas	0.0148	0.0155	-0.0156	0.0452
B_xi_1	gas	0.9800	0.0015	0.9771	0.9829
gamma	positive_static	0.1667	0.0111	0.1449	0.1886
zeta	positive_static	8.4733	0.0004	8.4726	8.4740
omega0	pi	-0.3912	0.0428	-0.4751	-0.3072
rho	pi	0.7675	0.0278	0.7129	0.8220
omega_y_1	pi	0.0376	0.0036	0.0304	0.0447

Note: Estimates from the optimizer used for the cached model. Standard errors and 95 % Wald CIs from the inverse Hessian; interpret with caution near optimizer bounds.

Model Comparison Tables

Section Overview

This section compares all models within each cadence (daily, monthly) side by side. Separate tables are provided for IS and OOS samples. The best value per metric column is highlighted in green. Red cells in coverage tables indicate rejection at the 5 % level.

Daily Models

Performance metrics and coverage tests for the daily cadence models.

Performance Metrics — Daily Models (IS)

Model	loglik	aic	bic	rmse	mad	crps
Daily ϕ-only	-5522.858	11065.716	11127.744	48.203	33.822	2.665
Daily ϕ, ξ	-5522.322	11070.644	11151.279	78.680	55.855	2.667

Note: Best value per metric highlighted in green. Lower is better for CRPS, RMSE, MAD, AIC, BIC; higher is better for log-likelihood.

Performance Metrics — Daily Models (OOS)

Model	rmse	mad	crps
Daily ϕ-only	43.992	34.015	2.670
Daily ϕ, ξ	72.621	56.365	2.673

Note: Best value per metric highlighted in green. Lower is better for CRPS, RMSE, and MAD.

95 % Coverage Tests — Daily (IS)

Model	violations	expected_violations	violation_rate	LR_uc	pvalue_uc	LR_cc	pvalue_cc	reject_uc_5pct	reject_cc_5pct
Daily ϕ-only	176.000	182.550	0.048	0.250	0.617	1.008	0.604	0.000	0.000
Daily ϕ, ξ	182.000	182.550	0.050	0.002	0.967	0.959	0.619	0.000	0.000

Note: Kupiec LR_uc tests unconditional 95 % coverage; Christoffersen LR_cc adds independence of violations. Red cells indicate rejection at the 5 % significance level.

95 % Coverage Tests — Daily (OOS)

Model	violations	expected_violations	violation_rate	LR_uc	pvalue_uc	LR_cc	pvalue_cc	reject_uc_5pct	reject_cc_5pct
Daily ϕ-only	37.000	36.550	0.051	0.006	0.939	0.015	0.993	0.000	0.000
Daily ϕ, ξ	37.000	36.550	0.051	0.006	0.939	0.015	0.993	0.000	0.000

Note: Kupiec LR_uc tests unconditional 95 % coverage; Christoffersen LR_cc adds independence of violations. Red cells indicate rejection at the 5 % significance level.

95 % Coverage Tests — Daily (Full)

Model	violations	expected_violations	violation_rate	LR_uc	pvalue_uc	LR_cc	pvalue_cc	reject_uc_5pct	reject_cc_5pct
Daily ϕ-only	213.000	219.100	0.049	0.180	0.671	0.875	0.645	0.000	0.000
Daily ϕ, ξ	219.000	219.100	0.050	0.000	0.994	0.874	0.646	0.000	0.000

Note: Kupiec LR_uc tests unconditional 95 % coverage; Christoffersen LR_cc adds independence of violations. Red cells indicate rejection at the 5 % significance level.

Monthly Models

Performance metrics and coverage tests for the monthly cadence models.

Conclusion

Summary of Findings

This report presents the first systematic comparison of two ZA-GAS specifications on the Belo Horizonte precipitation time series at daily and monthly cadences.

Daily OOS — lowest CRPS: Daily ϕ-only (2.6698).

Daily OOS — lowest RMSE: Daily ϕ-only (43.9918).

Coverage Diagnostics

Full-sample 95 % coverage tests do not reject for any model at the 5 % significance level.

Calibration (PIT)

PIT uniformity is not rejected for any model at the 5 % level, suggesting adequate probabilistic calibration.

Next Steps

Extend the analysis to all Brazilian weather stations to assess spatial consistency.

Compare ZA-GAS against benchmark models (climatological quantiles, seasonal ARIMA) on the same evaluation metrics.

Investigate whether ξ dynamics consistently improve calibration in high-rainfall regimes.

Explore alternative lag sets or seasonal parameterisations.

References

Creal, D., Koopman, S. J., & Lucas, A. (2012). Generalized autoregressive score models with applications. Journal of Applied Econometrics, 28(5), 777–795.

Kupiec, P. H. (1995). Techniques for verifying the accuracy of risk measurement models. Journal of Derivatives, 3, 73–84.

Christoffersen, P. F. (1998). Evaluating interval forecasts. International Economic Review, 39(4), 841–862.